Integrating Multitemporal SAR and Optical Information for Missing Optical Imagery Generation

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***Abstract*— Cloud cover and long revisit cycle of satellites can cause gaps in optical images and pose a significant obstacle to the consistency of Earth observation missions. Recently, synthetic aperture radar (SAR)-to-optical image translation (S2OIT) has become an emerging approach to reconstruct the missing information of optical remote sensing images. However, the previous studies ignored the mechanism difference between SAR and optical data and produced color distortion, image blurriness, and texture detail loss in the generated optical images. To tackle these challenges, we propose a multitemporal S2OIT network (MTS2ONet) for high-quality optical image generation. The proposed model comprises two subnetworks: change feature extraction subnetwork (Change_Extractor) and the S2OIT subnetwork (S2O_Translator). The first subnetwork is tasked with extracting change features from SAR images captured at dates T and $T + 1$, and then translating them from the SAR domain to the optical domain. Subsequently, the S2O_Translator integrates the optical image at date $T + 1$ with the change features extracted by the Change_Extractor to generate the optical image at date T. In addition, we produce a dual-temporal SAR–optical dataset called DTSEN1-2 for model evaluation. Experiments on the DTSEN1-2 dataset reveal that our method is superior to the state-of-the-art (SOTA) methods with the metrics peak-signal-to-noise ratio (PSNR; 36.0435), structural similarity index measure (SSIM; 0.9896), learned perceptual image patch similarity (LPIPS; 0.0443), and root mean square error (RMSE; 0.0174) and exhibits preferable results in visual effects. Our dataset and codes can be accessed via the following link: https://github.com/hopeupup/MTS2ONet.**

***Index Terms*— Deep learning, image reconstruction, multitemporal remote sensing, synthetic aperture radar (SAR)-to-optical image translation (S2OIT).**

I. Introduction

OPTICAL remote sensing images, exemplified by Sentinel-2 [1] and Landsat 8 [2], offer notable benefits including high resolution, obvious visual features, and strong interpretability. They have seen extensive utilization in the realms of image processing [3], [4], [5], [6], crop-type mapping [7], [8], [9], and ecological and environmental studies [10], [11], [12]. However, clouds usually cause serious interference in optical imaging systems. An analysis by the Moderate Resolution Imaging Spectroradiometer (MODIS) indicated that cloud coverage spans roughly 67% of the Earth's surface [13]. In addition, the revisit cycle of satellites can introduce critical data gaps and present substantial challenges in long-term optical remote sensing observations, such as wetland monitoring [14], landslide detection [15], and flood monitoring [16].

In the past two decades, numerous research endeavors have been dedicated to tackling the cloud cover of optical remote sensing images [17], [18], [19]. For example, Li et al. [20] introduced two expanded multitemporal dictionary learning algorithms [multitemporal beta process factor analysis (MT-BPFA) and multitemporal K singular value decomposition (MT-KSVD)] for recovering the quantitative products of optical and infrared remote sensing that are contaminated by large-area thick clouds and the consequent cloud shadows. Liu et al. [21] proposed a method for thick cloud removal under land cover changes using multisource satellite imagery and a spatiotemporal attention network. These methods lie in employing cloud-free images or bands from the reference phase to reconstruct the absent information in the target phase [22]. Several studies have applied synthetic aperture radar (SAR) data as auxiliary data to reconstruct higher quality optical imagery [23], [24]. However, these methods become limited when optical remote sensing images consistently encounter a high proportion of cloud cover.

The longer wavelength of SAR images, grants them day-and-night observation and cloud-penetration capabilities [25]. Recently, how to reasonably utilize SAR image information and translate it into optical image information has become

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a research hotspot. Zhang and Xu [26] applied an integrated fusion of intensity-hue-saturation and wavelet transform for SAR-to-optical image translation (S2OIT). Huang et al. [27] proposed a method based on sparse representation using low-resolution MODIS imagery and SAR imagery as auxiliary information to reconstruct Landsat imagery. With advancements in computing capabilities, deep learning has become the mainstream method in remote sensing. In particular, after the groundbreaking work of Goodfellow et al. [28], who unveiled the concept of the generative adversarial network (GAN) in 2014, there has been a notable surge in the adoption of GANs across image processing tasks, such as cloud removal [23], [29], [30], super-resolution [31], [32], [33], and image classification [34], [35], [36]. The emergence of GAN models has provided new insights into S2OIT [37], [38], [39]. Currently, the majority of GAN-based S2OIT methods are supervised, such as Pix2pix [40] and Pix2pixHD [41], requiring many paired training samples of SAR and optical imageries. In recent years, there has been the emergence of several datasets pairing SAR and optical imageries. Sebastianelli et al. [42] have developed a multimodal dataset pairing Sentinel-1 and Sentinel-2 images randomly distributed on the Earth's surface. Ebel et al. [43] have proposed a novel multimodal and multitemporal dataset called SEN12MS-CR-TS to tackle the challenge of optical satellite image reconstruction and cloud removal. Wang et al. [44] have produced a large-scale multimodal, multitemporal dataset for self-supervised learning in Earth observations, known as SSL4EO-S12. However, many optical images in these datasets are cloud covered, making them unsuitable for direct use in S2OIT tasks. Furthermore, the aforementioned methods rely solely on SAR image information and directly translate it into optical images, disregarding the fundamental differences in the physical representations of SAR and optical imageries. As a result, the translated images often suffer from color distortion and detail loss. Our method is designed to address these challenges.

In this article, we propose a GAN-based model named the multitemporal S2OIT network (MTS2ONet) for high-quality image translation. The generator of MTS2ONet consists of a change feature extraction subnetwork (Change_Extractor) and an S2OIT subnetwork (S2O_Translator). Both of these subnetworks combine residual blocks with attention mechanisms to enhance feature extraction capabilities. The Change_Extractor is responsible for converting change features of the same land cover from the SAR domain to the optical domain. The S2O_Translator integrates the features of the optical imagery at date $T + 1$ with the change features generated by the Change_Extractor and transforms them into an optical imagery at date T. The discriminator in the MTS2ONet is a multiscale discriminator capable of comprehensively capturing image details and structural information, thereby enriching the feature representation of the generated optical imagery. In addition, we produce a dual-temporal SAR–optical dataset (DTSEN1-2) to train and test MTS2ONet.

The summarization of our work's contribution is as follows.

1) We propose a GAN-based S2OIT network named MTS2ONet. The MTS2ONet model accepts SAR and optical imageries at date $T + 1$ and SAR imagery at date T as input to generate optical imagery at date T. The results show improvements in common issues encountered in S2OIT tasks, such as topographic deformation, spectral distortion, and color distortion.
2) In the MTS2ONet generator framework, we design a SAR–optical change feature extraction subnetwork called Change_Extractor and introduce a local loss to generate clearer and more accurate change features. Given the significant disparity between SAR features and optical features, directly combining these features can lead to unsatisfactory results. Therefore, the Change_Extractor subnetwork is responsible for translating the extracted change features from the SAR domain to the optical domain.
3) The advantage of our approach is to effectively integrate the temporal, spatial, and spectral information of multimodal and multitemporal imagery to reconstruct the whole imagery at the predicted date. Compared with state-of-the-art (SOTA) methods, our approach achieves the highest values on the metrics peak-signal-to-noise ratio (PSNR; 36.0435), structural similarity index measure (SSIM; 0.9896), learned perceptual image patch similarity (LPIPS; 0.0443), and root mean square error (RMSE; 0.0174).
4) We produce a DTSEN1-2 and utilize this dataset for assessing the outcomes of the suggested method. The DTSEN1-2 dataset consists of 4000 pairs of high-quality cloud-free dual-temporal SAR–optical data. Each pair of SAR–optical data includes four images: SAR and optical imageries at date T and SAR and optical imageries at date $T + 1$.

The subsequent sections in this article are structured in the following manner. The existing methods for optical imagery missing information reconstruction are reviewed in Section II. The architecture of the MTS2ONet is detailed in Section III. The details of the DTSEN1-2 dataset created for S2OIT tasks are provided in Section IV. Section V showcases the experimental results and analysis derived from comparative and ablation studies. In Section VI, we summarize our conclusion and outline future prospects.

II. Related Works

A. Cloud Removal Method

Traditional methods employ mathematical approaches [20], [45], [46] and physical models [47], [48], [49] to remove clouds and subsequently recover information that is partially obscured by clouds. For example, Xu et al. [46] effectively removed thin clouds present in optical imagery by implementing a noise-adjusted principal component transformation model (CR-NAPCT). Lv et al. [48] developed an algorithm founded on empirical and radiative transfer models to remove thin clouds in the visible spectrum and then recover ground features in cloud-covered regions. However, these models are typically designed for specific environmental conditions and may not be capable of handling various types of clouds. In particular, they face significant challenges in recovering optical images where most regions are cloud-contaminated or images

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containing thick clouds. With the advancement of deep learning, an increasing number of researchers are using data-driven methods for addressing cloud cover in optical remote sensing images [50], [51] to address several constraints found in traditional methods. Zheng et al. [29] have employed U-Net and GAN for single image cloud removal. Zhao et al. [30] have utilized multimodal GAN with adversarial consistency loss for cloud removal. Zhang et al. [52] have introduced a unified spatiotemporal spectral deep convolutional neural network model, STS-CNN, which can recover various types of missing information. However, the deep learning methods for cloud removal cause some spectral distortion and blurring when the thick clouds comprise a high ratio.

Due to the immunity of SAR imagery against cloud cover and adverse weather conditions, it can serve as auxiliary data to reconstruct optical imagery, where information are missing due to cloud cover or weather-related issues. For instance, Meraner et al. [53] introduced a cloud removal method for single-temporal Sentinel-2 satellite images based on a deep residual neural network (Dsen2-CR) alongside SAR–optical data. However, in cases where the optical image contains complex terrain or very thick cloud cover, the Dsen2-CR model cannot provide accurate reconstruction results. Sebastianelli et al. [23] proposed a novel multimodal method [pixel-level merging of intermediate feature maps (PLFM)] for cloud removal using temporal sequences of SAR and optical images. In this work, the translation from SAR-to-optical images is based on the pixelwise restoration of the whole scene. In addition, the proposed PLFM method reaches a very high PSNR (39.312 with a cloud cover of less than 0.2 and 37.247 with a cloud cover comprised between 0.2 and 0.4), while the accuracy decreases with a further increase of cloud cover.

B. SAR-to-Optical Image Translation

S2OIT methods directly convert SAR data into optical data without cloudy optical information, providing a new perspective for optical image generation. Existing S2OIT methods have been largely inspired by GANs and their variants, such as conditional GANs (CGAN) [54], Pix2pix, and CycleGAN [55]. For example, Schmitt et al. [56] explored the potential of using Pix2pix to simulate Sentinel-2 images from Sentinel-1 data. Hwang et al. [57] employed CycleGAN and simultaneously utilized SSIM loss and visual geometry group (VGG) perceptual loss to achieve S2OIT with richer structural information. Lin et al. [58] introduced a novel GAN model [post hoc attention (PHA) + residual hadamard production (RHP)], which incorporated a spatial attention mechanism to guide the generator's translation process. However, Pix2pix, CycleGAN, and PHA + RHP were not specifically designed for S2OIT tasks, and they overlook the notable disparities between SAR and optical imageries, leading to terrain deformations and spectral distortions in the translated optical imagery. Noa Turnes et al. [59] introduced an improved model called Atrous CGAN for S2OIT tasks. This model enhances fine details in the result images by leveraging multiscale spatial contextual features. Wang et al. [60] introduced a parallel GAN (Parallel-GAN) for S2OIT tasks. It effectively converts SAR images to optical images by introducing hierarchical latent optical features. Although these studies have made improvements to CGANs, the translated optical images still suffer from speckle noise and color distortion. Bai et al. [61] proposed a generative model (conditional diffusion) based on the diffusion process for S2OIT tasks. However, the generated images often exhibit color distortion due to the lack of color supervision during training. He and Yokoya [62] employed a CGAN with a residual architecture (MTcGAN) for optical image simulation from multitemporal SAR–optical data. The results indicate that optical imagery generated from multitemporal SAR–optical data is superior to that of imagery generated from single SAR data. This approach incorporates multitemporal SAR–optical information, bringing new ideas to S2OIT tasks. However, the MTcGAN model directly uses multitemporal SAR–optical data as input, overlooking the feature differences between multimodal data and the temporal differences within the same modality. As a result, the optical images generated by the MTcGAN model fail to accurately depict the change features of the same land.

To overcome the aforementioned challenges, we proposed a new GAN-based S2OIT model named MTS2ONet, which integrates the temporal, spatial, and spectral information from multimodal and multitemporal imagery to reconstruct missing optical images. The proposed model comprehensively considers the temporal variations in optical data as well as the differences between SAR and optical data, making the generated optical data more realistic. In addition, a cloud-free DTSEN1-2 is first established to satisfy the need for S2OIT tasks. This dataset encompasses diverse land use scenarios, ensuring the robust training of a widely applicable S2OIT model.

III. Method

A. Structure of MTS2ONet

The overall architecture of MTS2ONet is illustrated in Fig. 1. MTS2ONet is primarily composed of a generator and a discriminator, with detailed explanations provided below.

1) Generator: The generator comprises two key components: the Change_Extractor and the S2O_Translator subnetworks. The Change_Extractor is employed to obtain the change information in the optical domain by inputting SAR images at date T and date $T + 1$. The Change_Extractor consists of two downsampling layers, two upsampling layers, and an output layer. Data pass through the downsampling layers, which reduce the feature size. This operation decreases the computational load of the model and simultaneously enlarges its receptive field, enabling it to capture more representative contextual information. The downsampling layers include a residual module, C_ResBlock, and a max pooling layer. The depiction of the C_ResBlock's structure can be seen from Fig. 2(a). The upsampling layers consist of a C_ResBlock and a transpose convolution layer, designed to progressively restore the extracted feature maps to their original size and construct changes in the optical domain. Considering that pooling operations in the downsampling layers can lead to information loss and subsequently affect the model perception of details and boundaries, we introduced skip-connection layers in the Change_Extractor to preserve

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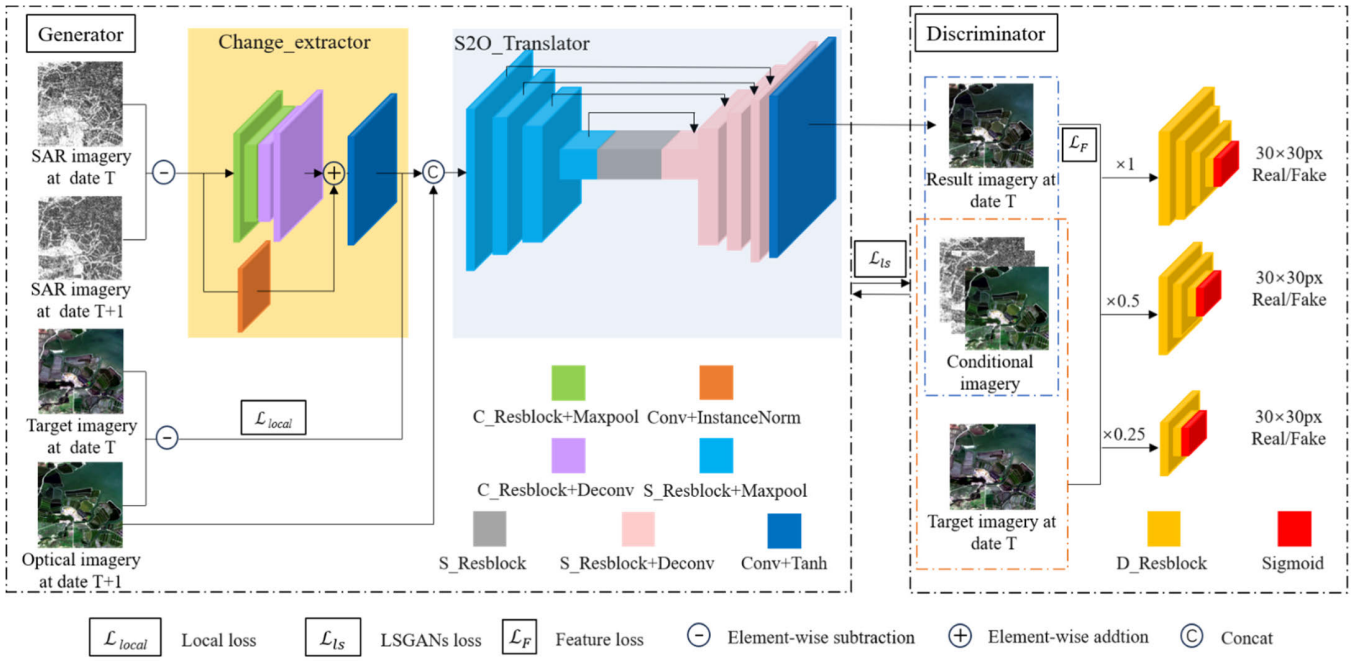


Fig. 1. Architecture of MTS2ONet.

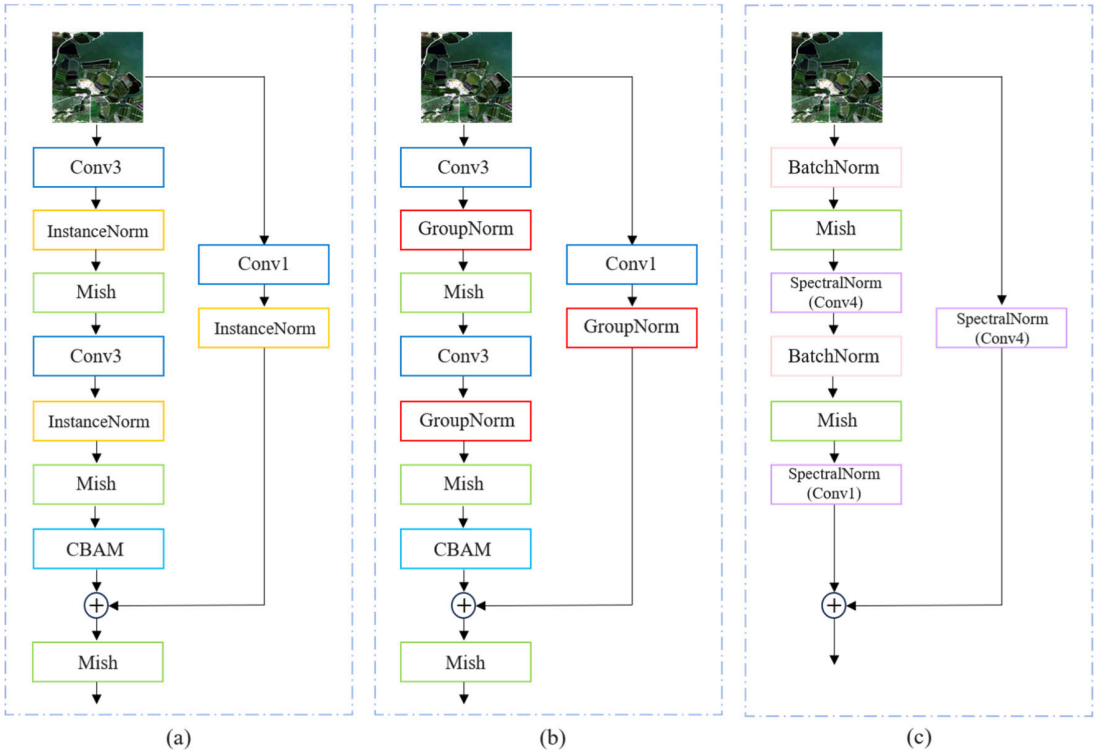


Fig. 2. Foundational residual blocks for the generator and the multiscale discriminator. (a) C_Resblock. (b) S_Resblock. (c) D_Resblock. (The Mish is an activation function).

the original information. Finally, the feature maps obtained through the upsampling layers, combined with those acquired from the skip connection layers, are input into an output layer consisting of a convolution and a Tanh activation function, generating change maps in the optical domain.

As depicted in Fig. 1, the S2O_Translator employs a U-net [63] architecture, which can simultaneously utilize local and global information to effectively extract target boundaries and fine structural features from the image, achieving high-quality image generation. The S2O_Translator is composed of both an encoder and a decoder. The primary role of the encoder is to capture the global features from change maps generated by the Change_Extractor and the optical imagery at date $T + 1$. The encoder is composed of four downsampling layers consisting of a residual module, S_ResBlock, and a Maxpool layer. The structure of the

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S_ResBlock is depicted in Fig. 2(b). The decoder is used to restore image details and comprises four upsampling layers, four skip connection layers, and one output layer. The upsampling layers consist of an S_ResBlock and a transpose convolution, and they are employed to integrate characteristics from both the encoder and decoder and restore the image size. The skip connection layers are used to connect features across various levels of the encoder and decoder. The output layer is composed of a convolution and a Tanh activation function and is employed to generate the optical images at date T.

As shown in Fig. 2, the choice of instance normalization for the C_Resblock is attributed to its success in image style transfer [64] and is expected to work well in transferring change information from the SAR domain to the optical domain in the Change_Extrator. Group normalization is chosen for the S_Resblock to ensure stable accuracy across a range of batch sizes [65]. Furthermore, we incorporated the convolutional block attention module (CBAM) [66] into both the C_ResBlock in the Change_Extractor and the S_ResBlock in the S2O_Translator. CBAM is a lightweight module that incorporates attention mechanisms for computer vision tasks. It introduces channel attention and spatial attention into deep convolutional neural networks. By applying weighting operations on the input feature maps in both channel and spatial dimensions, it computes weight coefficients, which are then multiplied with the feature maps to produce new feature maps. The CBAM module helps the network adaptively select and adjust the importance of features, enhancing its perceptual and feature extraction capabilities.

2) Discriminator: The role of the discriminator is to differentiate between the real and generated images, assessing the authenticity of the images. During the model training process, the quality of the generated images is consistently enhanced by the generator, bringing them closer to the target images. The discriminator continuously enhances its discriminatory ability, striving to differentiate between the generated and target images. A zero-sum game is conducted, resulting in higher quality generated images.

Inspired by the Patch-GAN architecture, a multiscale discriminator was developed, as depicted in Fig. 1. It consists of three different structured subdiscriminators, with each one being a regular binary classifier composed of several residual modules (D_ResBlock). Fig. 2(c) illustrates the architecture of the D_ResBlock. One challenge facing GAN models is instability during training. In response to this concern, spectral normalization was integrated into the D_Resblock to stabilize the multiscale discriminator's training process, resulting in the generation of superior-quality images [67]. The optical and SAR images at date $T+1$, along with the SAR image at date T, are regarded as conditions for guiding the training process. The inputs of the multiscale discriminator are the conditional images and the generated images at date T or the real images. Following the ultimate sigmoid activation, each discriminator transforms the input images into a probability map comprising 30×30 pixels for the categories of falseness and truthfulness. The multiscale discriminator enables image details and structural information to be more comprehensively captured, resulting in generated images with richer representations. To train three subdiscriminators with distinct architectures, the input images for the multiscale discriminator are downsampled into two varied scales with factors of 0.5 and 0.25.

B. Loss Functions

The efficacy of image generation significantly hinges on the choice of loss functions. Least squares generative adversarial networks (LSGANs) designed novel loss functions to produce superior quality images in comparison with traditional GANs and exhibit greater stability during the training process [68]. Motivated by these advantages, we integrated the loss functions utilized by LSGANs into the design of MTS2ONet. Given G denotes the generator and D denotes the discriminator. SAR^T and SAR^{T+1} represent the SAR data captured at date T and $T+1$, respectively, while Opt^T and Opt^{T+1} are used to denote the optical data at these same intervals. The objective function of G and D can be defined as follows, respectively:

$$\min_D \mathcal{L}_{\mathrm{ls}D}(D) = \frac{1}{2}\Big(\big(D\big(\mathrm{Opt}^T\big)-1\big)^2 + \big(D\big(G\big(\mathrm{SAR}^T, \mathrm{SAR}^{T+1}, \mathrm{Opt}^{T+1}\big)\big)\big)^2\Big) \tag{1}$$

$$\min_G \mathcal{L}_{\mathrm{ls}G}(G) = \frac{1}{2}\big(D\big(G\big(\mathrm{SAR}^T, \mathrm{SAR}^{T+1}, \mathrm{Opt}^{T+1}\big)\big)-1\big)^2. \tag{2}$$

To promote the generator, additional two losses are introduced. The first one is $\mathcal{L}_{\mathrm{Feature}}$, which is used to measure the consistency of the generated image and the real image. It is a composite of VGG loss ($\mathcal{L}_{\mathrm{R,VGG}}$), spectral angle loss ($\mathcal{L}_{\mathrm{Spectrum}}$), and visual loss ($\mathcal{L}_{\mathrm{MS\text{-}SSIM}}$) [69]. Furthermore, we proposed a local loss ($\mathcal{L}_{\mathrm{local_loss}}$) for the Change_Extractor to evaluate the transfer performance of the changing area from the SAR domain to the optical domain. $\mathcal{L}_{\mathrm{local}}$ also consists of VGG loss, spectral angle loss, and visual loss, while its inputs are the output of the Change_Extractor subnetwork and the difference map between Opt^T and Opt^{T+1}. It is worth noting that the target image at date T (Opt^T) only participates in the computation of $\mathcal{L}_{\mathrm{local}}$ during model training and is not employed in the validation and testing processes. Thus, the overall losses of the generator can be mathematically represented by the following equation:

$$\begin{aligned}\mathcal{L}_{\mathrm{total}G} = {} & \mathcal{L}_{\mathrm{ls}G}\big(\mathrm{SAR}^T, \mathrm{SAR}^{T+1}, \mathrm{Opt}^{T+1}\big) \\ & + \varepsilon\mathcal{L}_{\mathrm{Feature}}\big(\mathrm{SAR}^T, \mathrm{SAR}^{T+1}, \mathrm{Opt}^{T+1}\big) \\ & + \delta\mathcal{L}_{\mathrm{local}}\Big(\mathrm{SAR}^T, \mathrm{SAR}^{T+1}, \mathrm{Opt}^T, \mathrm{Opt}^{T+1}\Big)\end{aligned} \tag{3}$$

$$\mathcal{L}_{\mathrm{Feature}} = \alpha\mathcal{L}_{\mathrm{R,VGG}} + \beta\mathcal{L}_{\mathrm{Spectum}} + \lambda\mathcal{L}_{\mathrm{MS-SSIM}} \tag{4}$$

where ε, δ, α, β, and λ are the weighting coefficients.

$\mathcal{L}_{\mathrm{R,VGG}}$ is a feature matching loss function based on the pretrained VGG19 network [70]. This function effectively captures image details and semantic differences in images, making the content and style of the generated imagery closer to

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those of the original imagery. $\mathcal{L}_{\mathrm{R,VGG}}$ is calculated as follows:

$$\mathcal{L}_{\mathrm{R,VGG}} = \sum_{i=1}^{5} \frac{1}{H_i W_i C_i} \|\phi_i(y) - \phi_i(\hat{y})\|_1 \tag{5}$$

where $\hat{y}$ and y denote the generated image and target image for $\mathcal{L}_{\mathrm{Feature}}$, respectively. As for $\mathcal{L}_{\mathrm{local}}$, $\hat{y}$ and y denote the output of Change_Extractor and the difference map between Opt^T and Opt^{T+1}. i is the order of the five layers in VGG19, namely, the 2nd, 7th, 12th, 21st, and 30th layers. Here, $\phi_i(*)$ represents the feature map of the ith layer in VGG19, C_i signifies the number of channels in the ith layer of the VGG19, and (H_i, W_i) denotes the dimensions of the feature map on each channel at the ith layer of the VGG19.

To mitigate spectral distortion, cosine similarity between the result and target imagery is employed to reduce and control the spectral angle loss ($\mathcal{L}_{\mathrm{Spectrum}}$). The definition of $\mathcal{L}_{\mathrm{Spectrum}}$ is provided in (4) as follows:

$$\mathcal{L}_{\mathrm{Spectum}} = I - \frac{\hat{y} \cdot y}{\|\hat{y}\| \|y\|} \tag{6}$$

where I represents a tensor with a value of 1.

The visual loss ($\mathcal{L}_{\mathrm{MS\text{-}SSIM}}$) can be used to assess the textural information of the generated images. We employ multiscale structural similarity (MS-SSIM) as the visual loss

$$\mathcal{L}_{\mathrm{MS-SSIM}} = I - \left[l_M(\hat{y}, y)\right]^{\alpha_M} \prod_{j=1}^{M} \left[c_j(\hat{y}, y)\right]^{\beta_j} \left[s_j(\hat{y}, y)\right]^{\gamma_j} \tag{7}$$

$$l_M(\hat{y}, y) = \frac{2\mu_{\hat{y}}\mu_y + C_1}{\mu_{\hat{y}}^2 + \mu_y^2 + C_1} \tag{8}$$

$$c_j(\hat{y}, y) = \frac{2\sigma_{\hat{y}}\sigma_y + C_2}{\sigma_{\hat{y}}^2 + \sigma_y^2 + C_2} \tag{9}$$

$$s_j(\hat{y}, y) = \frac{\sigma_{\hat{y}y} + C_3}{\sigma_{\hat{y}}\sigma_y + C_3} \tag{10}$$

where α_M, β_j, and γ_j are the corresponding weighting coefficients; $\mu_{\hat{y}}$ and μ_y are the mean of images $\hat{y}$ and y, respectively; $\sigma_{\hat{y}}$ and σ_y are the variance of images $\hat{y}$ and y, respectively; $\sigma_{\hat{y}y}$ is the covariance of images $\hat{y}$ and y; and C_1, C_2, and C_3 are small constants. The calculations for contrast comparison $c_j(\hat{y}, y)$ and structure comparison $s_j(\hat{y}, y)$ are conducted at the jth scale, whereas the luminance comparison $l_M(\hat{y}, y)$ is specifically computed only at the highest scale M.

C. Training Setting

1) Evaluation Indicators: We selected four mainstream quantitative indices to evaluate the performances of different models, including the SSIM [71], PSNR [72], RMSE [73], and LPIPS metric [74].

These metrics are calculated as follows:

$$\mathrm{PSNR}(x, y) = \frac{10 \log_{10} \mathrm{Max}\left(y_{c,h,w}\right)}{\|x_{c,h,w} - y_{c,h,w}\|_2} \tag{11}$$

where x and y denote the generated and target imagery respectively, and $x_{c,h,w}$ and $y_{c,h,w}$ represent their corresponding pixel values

$$\mathrm{RMSE} = \sqrt{\frac{1}{C \cdot H \cdot W} \sum_{c=h=w=1}^{C \cdot H \cdot W} \left(y_{c,h,w} - x_{c,h,w}\right)^2} \tag{12}$$

TABLE I
EXPERIMENTAL HARDWARE AND SOFTWARE ENVIRONMENTS

Hardware	RAM	CPU	GPU
	64G	AMD Ryzen 9 5950X	RTX 3080Ti
Software	Python	CUDA	Pytorch
	3.8	11.3	1.12

where C, H, and W represent the number of channels, height, and weight of the image, respectively.

SSIM is a widely employed metric for assessing the similarity between the result and target imagery with respect to their brightness, contrast, and structure. As the SSIM value approaches 1, it signifies a higher degree of resemblance between the two images. SSIM is calculated as follows:

$$\mathrm{SSIM}(x, y) = \frac{\left(2x_{c,h,w}y_{c,h,w} + c_1\right)\left(2\sigma_{xy} + c_2\right)}{\left(\mu_x^2 + \mu_y^2 + c_1\right)\left(\sigma_x^2 + \sigma_y^2 + c_2\right)} \tag{13}$$

where μ_x and μ_y indicate the mean pixel values of x and y, respectively. σ_x and σ_y correspond to the standard deviations of x and y, respectively. σ_{xy} denotes the covariance between x and y. c_1 and c_2 represent the fixed constants.

LPIPS serves as a valuable metric for assessing the subjective perceptual disparity between the result and target imagery. A lower LPIPS indicates a higher similarity between the two images. LPIPS is calculated as follows:

$$\mathrm{LPIPS}\left(y, y^*\right) = \sum_{l} \frac{1}{H_l W_l} \sum_{h,w} \left\| w_l \odot \left(a_{hw}^l - a_{hw}^{*l}\right)\right\|_2^2 \tag{14}$$

where a_{hw}^l and a_{hw}^{*l} represent the feature maps of y and y^* at the lth layer of the evaluation network, respectively, and w_l denotes the cosine distance between feature maps.

2) Implementation Details: We trained, validated, and tested the MTS2ONet model on the DTSEN1-2 dataset. The dataset was randomly divided into training, validation, and test sets in an 8:1:1 ratio. We also conducted a comparison between MTS2ONet and seven advanced methods, including Pix2pix [40], CycleGAN [55], Parallel-GAN [60], PHA + RHP [58], conditional diffusion [61], Dsen2-CR [53], and MTcGAN [62]. Since Dsen2-CR is a classic cloud removal method, we replaced its input from a cloudy optical image with the same input as MTS2ONet for a fair comparison. The experimental setup involved PyTorch and an NVIDIA 3080Ti GPU. A detailed description of the experimental hardware and software environments is shown in Table I. The models were optimized using the Adam optimizer with momentum terms set to (0.9, 0.999). All models were trained for 200 epochs with a batch size of 4 and a learning rate of 0.001. In (3), $\delta = \varepsilon = 0.1$; in (4), $\alpha = \beta = \gamma = 1$; in (7)–(10), we set $M = 5$ as in [75], while other parameters follow the settings described in [69]; and in (13), $c_1 = 1e^{-4}$ and $c_2 = 9e^{-4}$ [76]. Table II displays the input and parameter size of the models. The generator model parameter count of MTS2ONet is 6.25% that of Parallel-GAN and 61.18% that of Pix2pix. In terms of discriminator model parameters, our MTS2ONet model uses a multiscale discriminator consisting

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TABLE II
INPUT AND PARAMETER SIZE OF DIFFERENT NETWORK MODELS

Methods	Input	Total number of params	
		Generators	Discriminators
Pix2pix [40]	SAR image at date T	54.412 M	2.767 M
CycleGAN [55]	SAR image at date T	22.754 M	5.529 M
Parallel-GAN [60]	SAR image at date T	532.200 M	11.157 M
PHA+RHP [58]	SAR image at date T	22.756 M	5.530 M
Conditional Diffusion [61]	SAR image at date T	124.054 M	\
Dsen2-CR [53]	SAR images at date T and T+1, optical image at date T+1	18.891 M	\
MTcGAN [62]	SAR images at date T and T+1, optical image at date T+1	11.383 M	2.772 M
MTS2ONet (Ours)	SAR images at date T and T+1, optical image at date T+1	33.288 M	24.365 M

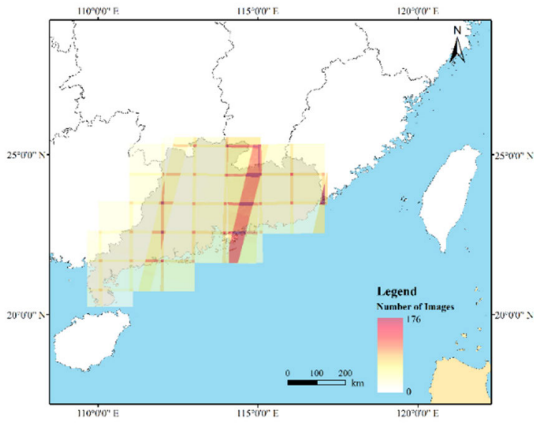


Fig. 3. Number of Sentinel-2 images with cloud cover below 20% in Guangdong Province in 2022.

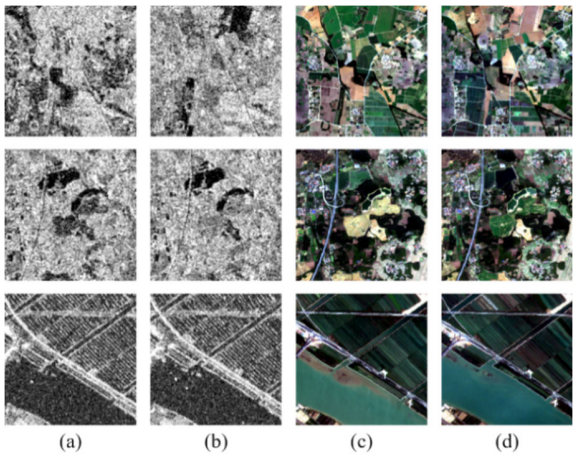


Fig. 4. Images from the DTSEN1-2 dataset. (a) SAR images at date T. (b) SAR images at date $T + 1$. (c) Optical images at date T. (d) Optical images at date $T + 1$.

of three different discriminators, so the parameter count of the discriminator in the MTS2ONet is larger than in other models.

IV. DTSEN1-2 DATASET

In this article, we generated a DTSEN1-2 for S2OIT tasks. The DTSEN1-2 dataset was produced on the Google Earth Engine (GEE) platform. The collection area of the DTSEN1-2 dataset is located in Guangdong Province, China. Guangdong Province frequently experiences cloudy and rainy weather conditions, resulting in a scarcity of distinct optical remote sensing images in most areas. As illustrated in Fig. 3, the number of Sentinel-2 images for Guangdong Province in 2022 is very limited. The proposed model shows great potential for reconstructing missing optical images in this region. Therefore, Guangdong Province is selected as the collection area to validate and apply the proposed model.

The collection area included the overlap regions of Sentinel-1 imagery and Sentinel-2 imagery from March 1, 2022 to March 10, 2022 and April 1, 2022 to April 10th, 2022, in Guangdong Province. The absolute difference in acquisition time between paired Sentinel-1 and Sentinel-2 images is ensured to be less than three days. To obtain paired dual-temporal SAR–optical data, the locations of each sampling point were randomly and uniformly extracted within the overlap region.

We acquired Sentinel-2 imagery at the 2-A level on the GEE platform. The RGB bands were selected as the optical remote sensing data in our dataset. These Sentinel-2 images have undergone preprocessing operations, including orthorectification, geometric correction, and atmospheric correction, resulting in more accurate color levels and reflectance data. The Sentinel-1 imagery dataset obtained on the GEE platform is the Level-1 ground range detected (GRD) product. It has undergone preprocessing operations, such as thermal noise removal, radiometric calibration, and terrain correction. We selected the single co-polarization [vertical transmit/horizontal receive (VH)] data from the interferometric wide swath (IW) mode as the SAR data for our dataset. The values are σ° backscatter coefficients that have been converted to decibel scale. The DTSEN1-2 dataset consists of a total of 4000 cloud-free paired dual-temporal SAR–optical data pairs. Each pair of SAR–optical data includes four images (Fig. 4): SAR and optical images at date T, as well as SAR and optical images at date $T + 1$. The date T and date $T + 1$ represent two periods: from March 1st to March 10th and from April 1st to April 10th. In total, the DTSEN1-2 dataset comprises 16 000 images, each with dimensions of 256 × 256 pixels.

V. RESULTS

A. Qualitative Evaluation

As shown in Figs. 5–8, we selected four different scenes, including farmland, mountain, water, and building,

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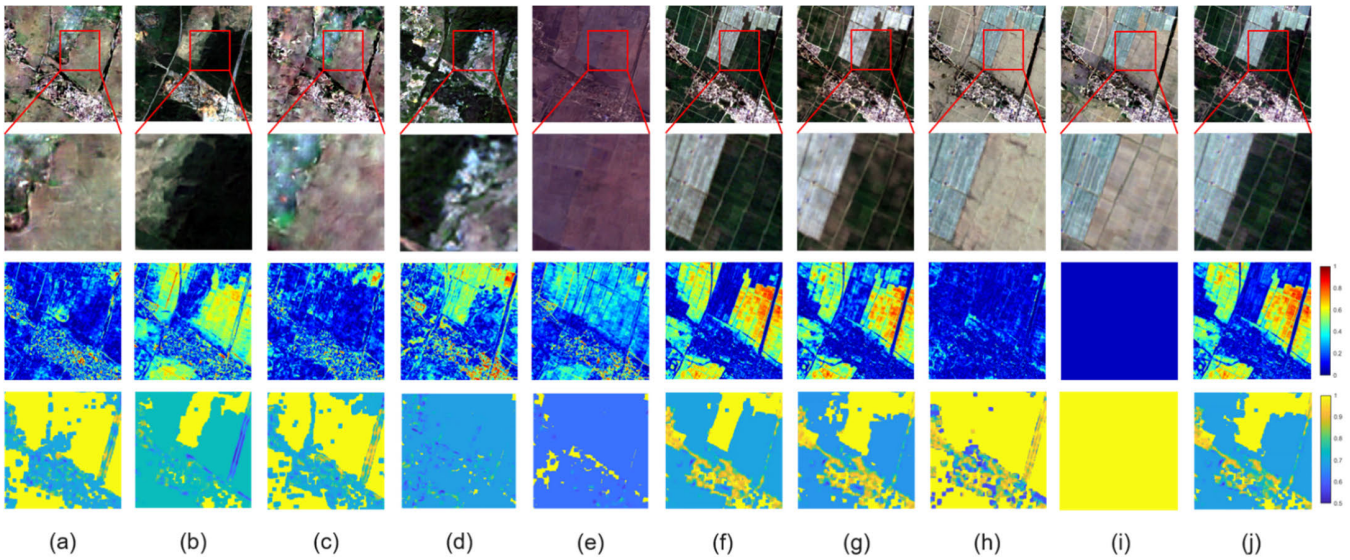


Fig. 5. Results of farmland scene. (a) Pix2pix. (b) CycleGAN. (c) Parallel-GAN. (d) PHA + RHP. (e) Conditional diffusion. (f) Dsen2-CR. (g) MTcGAN. (h) MTS2ONet (ours). (i) Target images at date T. (j) Optical images at date $T + 1$.

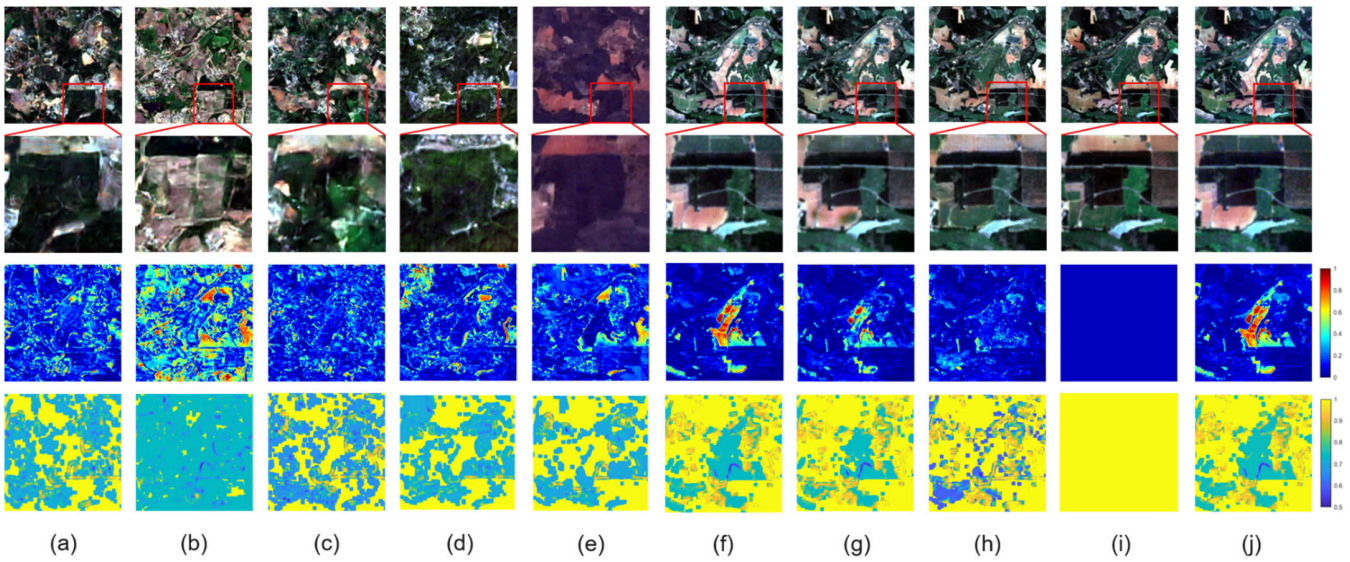


Fig. 6. Results of mountain scene. (a) Pix2pix. (b) CycleGAN. (c) Parallel-GAN. (d) PHA + RHP. (e) Conditional diffusion. (f) Dsen2-CR. (g) MTcGAN. (h) MTS2ONet (ours). (i) Target images at date T. (j) Optical images at date $T + 1$.

to qualitatively evaluate the quality of images generated by various models. The first row depicts the generated images from each model and the corresponding real image. The second row depicts the zoomed-in view detail of these images. The third row depicts the residual maps from comparing the generated images with the target images. The fourth row displays the similarity maps. The closer the values of the residual map are to 0, the bluer the map appears, and the higher the quality of the image is; the closer the values of the similarity map are to 1, the more yellow the map appears, and the higher the quality of the image is.

In the farmland scene (Fig. 5), the images generated by Pix2pix and Parallel-GAN are consistent in color with the real images, whereas the colors of the images generated by CycleGAN, PHA + RHP, and conditional diffusion are entirely different from those of the real images. In addition, all five models suffer from issues, such as color distortion, missing texture details, and unclear object boundaries in the imagery. On the contrary, the images generated by Dsen2-CR, MTcGAN, and MTS2ONet exhibit a high degree of structural similarity to target images. Notably, the result images of the MTS2ONet model closely resemble the target images in terms of both content and style.

In the mountain scene, depicted in Fig. 6, the images generated by Pix2pix, CycleGAN, Parallel-GAN, PHA + RHP, and conditional diffusion exhibit poor overall visual quality and lack local details. Furthermore, the residual maps and similarity maps illustrate that the result images differ significantly from the target images. Conversely, the images generated by Dsen2-CR, MTcGAN, and MTS2ONet demonstrate good structural similarity and clear object boundaries. In particular, the proposed model MTS2ONet generates

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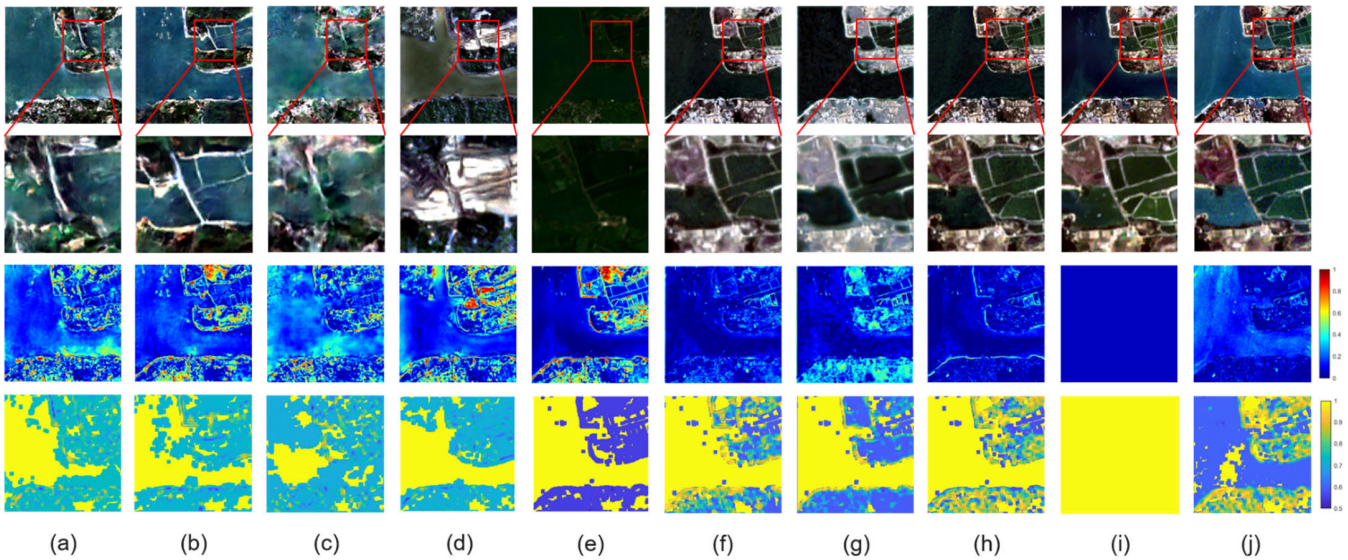


Fig. 7. Results of the water scene. (a) Pix2pix. (b) CycleGAN. (c) Parallel-GAN. (d) PHA + RHP. (e) Conditional diffusion. (f) Dsen2-CR. (g) MTcGAN. (h) MTS2ONet (ours). (i) Target images at date T. (j) Optical images at date $T + 1$.

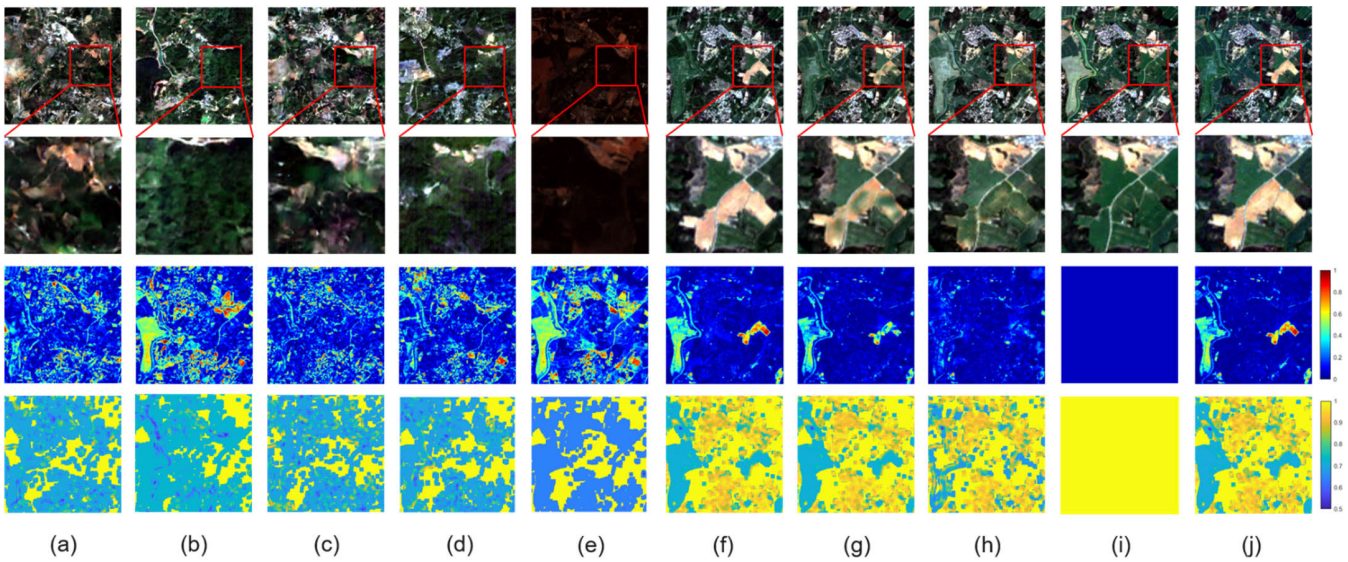


Fig. 8. Results of the building scene. (a) Pix2pix. (b) CycleGAN. (c) Parallel-GAN. (d) PHA + RHP. (e) Conditional diffusion. (f) Dsen2-CR. (g) MTcGAN. (h) MTS2ONet (ours). (i) Target images at date T. (j) Optical images at date $T + 1$.

images that effectively portray the change in land in different phases. In contrast, Dsen2-CR and MTcGAN tend to emphasize optical image features at date $T + 1$ or the fusion of features between date T and date $T + 1$.

In the water scene, depicted in Fig. 7, the images generated by the Pix2pix, CycleGAN, Parallel-GAN, PHA + RHP, and conditional diffusion models tend to exhibit color distortion. As shown by the zoomed-in view detail in the second row, these models also neglect texture details and sometimes incorrectly translate bare land into mountainous terrain. In contrast, the optical images generated by Dsen2-CR, MTcGAN, and MTS2ONet are consistent with the target images in overall style. However, when examining the zoomed-in-view detail in the second row, the result images of Dsen2-CR and MTcGAN have lighter colors and exhibit image blurriness. Moreover, the corresponding residual maps and similarity maps demonstrate that the features of bare land near the river valleys and buildings in the images generated by Dsen2-CR and MTcGAN differ from those in the target images.

For the building scene, depicted in Fig. 8, Pix2pix, CycleGAN, Parallel-GAN, PHA + RHP, and conditional diffusion incorrectly translate farmland into mountainous terrain, as illustrated by the zoomed-in-view detail in the second row. In terms of the overall visual results shown in the first row, Dsen2-CR, MTcGAN, and the proposed MTS2ONet all depict the buildings well. However, examining the zoomed-in-view detail in the second row reveals that the result images of MTS2ONet display roads and other objects more clearly, better aligning with the target images.

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TABLE III
COMPARISON OF RESULTS WITH QUANTITATIVE EVALUATION OF PSNR, SSIM, LPIPS, AND RMSE BY DIFFERENT MODELS ON OUR TEST SET

Methods	PSNR↑	SSIM↑	LPIPS↓	RMSE↓
Pix2pix [40]	26.7167	0.9423	0.2247	0.0480
CycleGAN [55]	24.4719	0.9025	0.2777	0.0612
Parallel-GAN [60]	28.1406	0.9555	0.1998	0.0411
PHA+RHP [58]	25.6907	0.9214	0.2631	0.0538
Conditional Diffusion [61]	28.3222	0.4175	0.4238	0.1437
Dsen2-CR [53]	34.9720	0.9877	0.0594	0.0195
MTcGAN [62]	34.5817	0.9859	0.0545	0.0205
MTS2ONet (Ours)	**36.0435**	**0.9896**	**0.0443**	**0.0174**

Figs. 5–8 illustrate that Pix2pix, CycleGAN, Parallel-GAN, PHA + RHP, and conditional diffusion, which use only single-temporal SAR images for S2OIT tasks, commonly suffer from issues, such as color distortion, image blurriness, and missing texture details. This is because they did not utilize the optical image at date $T + 1$. SAR images provide detailed surface information of objects, but optical images may have varying appearances under different weather and lighting conditions. Directly translating a SAR image to an optical image without prior optical information is very difficult. Dsen2-CR, MTcGAN, and MTS2ONet utilize multitemporal SAR and optical images to produce images that are more aligned with the target images in terms of image style, texture details, and other aspects, demonstrating the effectiveness and advantages of feature fusion between SAR and optical data in S2OIT. However, Dsen2-CR and MTcGAN do not effectively capture the changes in surface characteristics at different times. This is because they directly fuse SAR and optical features, ignoring their imaging mechanism and temporal aspect differences, resulting in the advantages of different modal data being underutilized. In contrast, MTS2ONet utilizes the Change_Extractor to transform the changing areas between two dates from the SAR domain to the optical domain to help the model generate a more real image. Thus, the results [column (h)] effectively represent the changes in surface characteristics and maintain a higher level of consistency with the target images.

B. Quantitative Evaluation

We applied four metrics to quantitatively evaluate the performance of the MTS2ONet model in comparison with other models. The results in Table III reveal that MTS2ONet achieved a PSNR of 36.0435, an SSIM of 0.9896, an LPIPS score of 0.0443, and an RMSE of 0.0174. MTS2ONet outperformed other models significantly in the PSNR and RMSE metrics, indicating that the result images of our model are closer to target images at the pixel level. MTS2ONet also outperformed the other models in the SSIM metric, suggesting that the result images have an overall structure closer to the target images. Notably, the results of MTS2ONet surpassed those of other models in the LPIPS metric, indicating that our model excels in human subjective visual perception and generates images that visually align more closely with human perceptual standards.

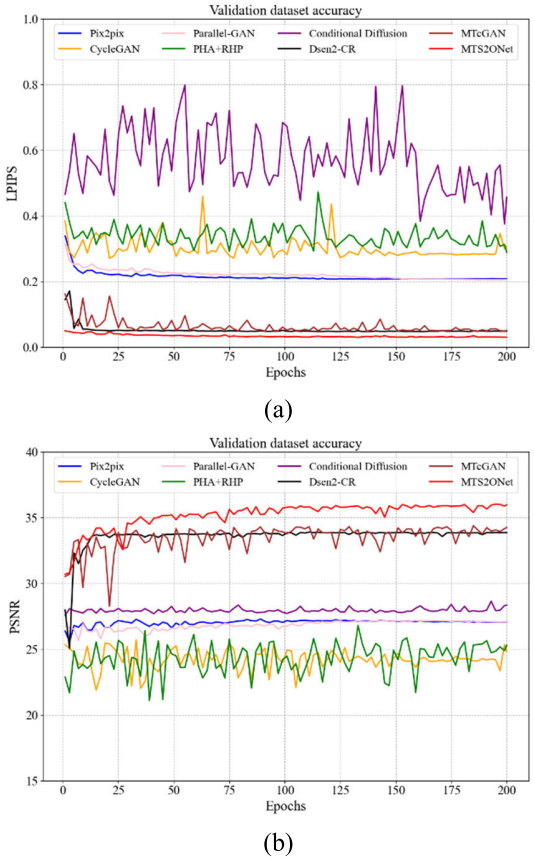


Fig. 9. Error variation during model validation on the validation dataset. (a) LPIPS of models on the validation dataset. (b) PSNRs of the models on the validation dataset.

Fig. 9 displays the variations of LPIPS and PSNR during the model training on the validation dataset. The proposed MTS2ONet achieves better image quality on the validation dataset than the other models. Furthermore, the models that integrate SAR and optical features, namely, Dsen2-CR, MTcGAN, and MTS2ONet, exhibit significantly higher PSNR and LPIPS values for images generated on the validation dataset than the models that utilize only single-temporal SAR images, namely, Pix2pix, CycleGAN, Parallel-GAN, PHA + RHP, and conditional diffusion. Moreover, MTS2ONet exhibits faster convergence and greater stability during training on the validation dataset than the other models.

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TABLE IV
ABLATION STUDY OF THE MTS2ONet STRUCTURE WITH FOUR EVALUATION METRICS ON OUR TEST SET

	PSNR↑	SSIM↑	LPIPS↓	RMSE↓
MTS2ONet without CBAM	35.5890	0.9882	0.0470	0.0186
MTS2ONet with single scale	35.7171	0.9892	0.0451	0.0181
MTS2ONet	**36.0435**	**0.9896**	**0.0443**	**0.0174**



Fig. 10. (a) MTS2ONet without CBAM. (b) MTS2ONet with a single scale. (c) MTS2ONet. (d) Target images at date T. (e) Optical images at date $T + 1$.

TABLE V
ABLATION STUDY OF THE MTS2ONet LOSS FUNCTION WITH FOUR EVALUATION METRICS ON OUR TEST SET

	PSNR↑	SSIM↑	LPIPS↓	RMSE↓
MTS2ONet without $\mathcal{L}_{local_loss}$	35.8157	0.9891	0.0467	0.0179
MTS2ONet without $\mathcal{L}_{\text{VGG}}$	35.1482	0.9882	0.0500	0.0192
MTS2ONet without $\mathcal{L}_{\text{Spectrum}}$	35.8045	0.9888	0.0456	0.0180
MTS2ONet without $\mathcal{L}_{\text{MS-SSIM}}$	35.7100	0.9887	0.0469	0.0181
MTS2ONet	**36.0435**	**0.9896**	**0.0443**	**0.0174**

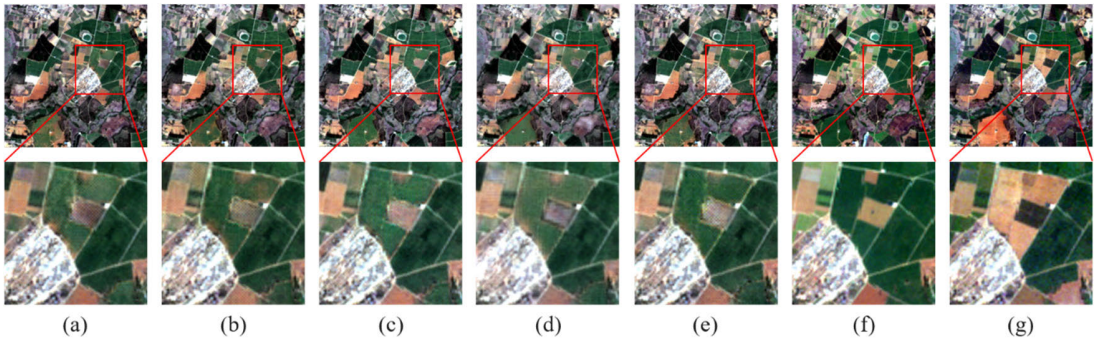


Fig. 11. (a) MTS2ONet without $\mathcal{L}_{\text{local_loss}}$. (b) MTS2ONet without $\mathcal{L}_{\text{Spectrum}}$. (c) MTS2ONet without $\mathcal{L}_{\text{MS-SSIM}}$. (d) MTS2ONet without L_{VGG}. (e) MTS2ONet. (f) Target images at date T. (g) Optical images at date $T + 1$.

C. Ablation Experiment

To validate the effectiveness of various components in MTS2ONet, we assessed the impacts of the CBAM module, $\mathcal{L}_{\text{local_loss}}$, $\mathcal{L}_{\text{VGG}}$, $\mathcal{L}_{\text{spectrum}}$, $\mathcal{L}_{\text{MS-SSIM}}$, and multiscale discriminator on the performance of MTS2ONet.

We first removed the CBAM module, which combines channel attention and spatial attention mechanisms to enhance the model's focus on key features. As evidenced in Table IV, the exclusion of the CBAM module results in a certain degree of degradation in PSNR, SSIM, LPIPS, and RMSE for the result images of the MTS2ONet. This illustrates the crucial role of the CBAM module in improving the accuracy and quality of image translation. Second, we modified the multiscale discriminator to be a single-scale discriminator to verify the effectiveness of the multiscale discriminator. Table IV indicates that the model with a multiscale discriminator produces higher quality images than the model with a single-scale discriminator in terms of PSNR, SSIM, LPIPS,

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and RMSE. The overall and locally magnified details of the images generated by MTS2ONet without the CBAM module, MTS2ONet with a single-scale discriminator, and complete MTS2ONet are illustrated in Fig. 10. The result images of the complete MTS2ONet are the closest to target images.

We also individually removed $\mathcal{L}_{\text{local_loss}}$, $\mathcal{L}_{\text{VGG}}$, $\mathcal{L}_{\text{spectrum}}$, and $\mathcal{L}_{\text{MS-SSIM}}$ to validate the rationality of their composite function. Table V indicates that when each of these loss functions is removed separately, the PSNR, SSIM, LPIPS, and RMSE of the images generated by MTS2ONet decline to varying degrees. This suggests that these loss functions are crucial in the training of the MTS2ONet, contributing to the enhancement of perceptual quality and structural similarity in generated images. Notably, the removal of $\mathcal{L}_{\text{VGG}}$ most significantly reduces performance. This is because VGG loss utilizes intermediate layer features from the VGG network to calculate the loss. These layers contain more semantic information, making it easier for the generated images to maintain consistency with target images in terms of objects, textures, and structures. The overall and locally magnified details of the images generated by MTS2ONet with different loss functions, as well as the original MTS2ONet, are depicted in Fig. 11. The result images of the MTS2ONet exhibit the highest similarity to target images.

VI. Conclusion

S2OIT is currently a popular research topic, but some challenges, such as color distortion, image blurriness, and the loss of texture details, have not yet been fully addressed. To overcome these challenges, we propose the MTS2ONet model, which integrates multitemporal SAR and optical information to facilitate the generation of high-quality optical images. The main features of the MTS2ONet model encompass two aspects. The Change_Extractor subnetwork is used to learn more effective mapping relationships to transform changes in the geographic features from the SAR domain to the optical domain. The S2O_Translator subnetwork fuses optical features at date $T + 1$ and change features generated by Change_Extractor, ensuring the generated optical images and the target images exhibit consistent texture details, color accuracy, and color saturation. We also produce a DTSEN1-2 and conduct extensive experiments on it. Through both quantitative evaluation and visual inspection, these experiments reveal that the MTS2ONet model achieves leading-edge performance and strong robustness in S2OIT tasks. The MTS2ONet model depends on the availability of cloud-free optical imagery at date $T + 1$. If such imagery is unavailable for a prolonged period, it will hinder the generation of missing optical imagery. In the subsequent work, our aim is to investigate a more efficient and lightweight model to enhance speed and quality in S2OIT tasks. In addition, we will also attempt research using SAR and optical paired datasets with a larger temporal gap and explore the applications of fully polarized SAR imagery and multiband optical imagery in S2OIT tasks.

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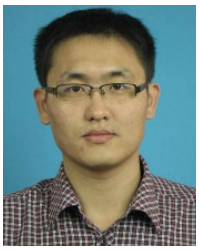
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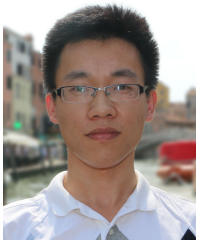
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